

CONFORMATIONAL NANO-SWITCH FOR FUNCTIONAL MULTIPLEXING BY CELL-SIGNALING RECEPTORS

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ABSTRACT

A central challenge in biology is determining how some receptors generate multiple, and sometimes even opposite, signals in cells. Here, we investigate the origin of multi-signaling features prevalent in life sciences with focus on nanoscale conformational switches triggered by Fas receptor-ligand interactions. The prototypical multiplexing receptor, Fas (also known as CD95) drives both pro-life and pro-death signalling pathways; the molecular mechanisms of which remain unknown. Through an interdisciplinary approach combining genetic code expansion technology for click chemistry on mammalian receptors and single-molecule fluorescence resonance energy transfer (smFRET) spectroscopy, we observe ligand-induced conformational re-arrangement of the Fas death domain, potential serving as switch for selective recruitment of cytosolic signaling complexes.